

Pythagorean Proof Gallery Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	pythagorean-proof-gallery

Big idea: Mathematical proof is not just a result; it is a standard for why a result must be true.

Overview

Students compare visual, algebraic, and verbal proofs of the Pythagorean theorem and decide what makes each convincing.

Student Prompt

Inspect Pythagorean Proof Gallery Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Pythagorean Proof Gallery Stimulus A	Student-facing stimulus 1: Visual rearrangement proof of $a^2 + b^2 = c^2$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Pythagorean Proof Gallery Stimulus B	Student-facing stimulus 2: Algebraic proof card plus a verbal explanation card of the same theorem. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Pythagorean Proof Gallery Reveal / Twist	Student-facing stimulus 3: Algebraic proof card plus a verbal explanation card of the same theorem. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical proof is not just a result; it is a standard for why a result must be true.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Pythagorean Proof Gallery Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Pythagorean Proof Gallery Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	What makes mathematical proof different from empirical confirmation?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how mathematical proof is not just a result; it is a standard for why a result must be true.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare three proof formats and do not identify one as best at the start.
- Ask students to rank proof cards using different criteria: certainty, explanation, elegance, and accessibility.

Ready-to-Use Examples

- Visual rearrangement proof of $a^2 + b^2 = c^2$.
- Algebraic proof card plus a verbal explanation card of the same theorem.

Suggested Timing

Stage	Teacher move	Watch for
1	Display three different proofs of $a^2 + b^2 = c^2$.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups annotate what each proof shows, assumes, and leaves unclear.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students rank proofs by convincingness, explanatory power, and elegance.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups compare whether their rankings changed across criteria.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why mathematical knowledge values proof rather than repeated measurement alone.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- A proof does more than persuade; it should show why the result must hold.
- Which proof gives certainty, and which proof gives understanding?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: focus on two proof cards.
- Challenge: students write a fourth proof explanation for younger students.

Extension Tasks

- Students compare a proof with many measured right triangles and explain why measurement is insufficient.
- Connect mathematical elegance to aesthetic judgement in the arts.

Related Activities

- Proof versus Pattern
- Axiom Switch Universe
- Calculator Trust Test